

Design Attributes of A Traditional DTV SoC

Design attribute	Approximate value
Die-size	≤70 mm ² in 65 nm
Gate count	≤25 million
Power dissipation	8-12W
Logic speed	300-400 MHz
32-bit DDR2/3 memory speed	≤1.066 Gbps
Host processor speed	800 MHz
Audio DSP, security and transport processor speed	400 MHz
2D/3D graphics engine speed	200 MHz
MIPS	500

encoders supporting new functionality, multi-view coding or distributed video coding, support for next-generation inputs like HDMI and LVDS ++, and standard and legacy interfaces. DTV OEMs would typically want to take advantage of advanced nodes like 28nm/22nm technology to achieve power and cost efficiencies during the

realisation of next-generation DTV media SoC.

Handling next-generation complex DTV SoC design complexities and managing time-to-market and cost pressures today pose a huge challenge for OEMs. Optimisation of system-modeling, verification, prototyping, emulation and silicon validation cycles of the SoC development can help DTV OEMs reduce time to market. A unified domain-aware modeling-verification-emulation-validation approach holds such a strong promise.

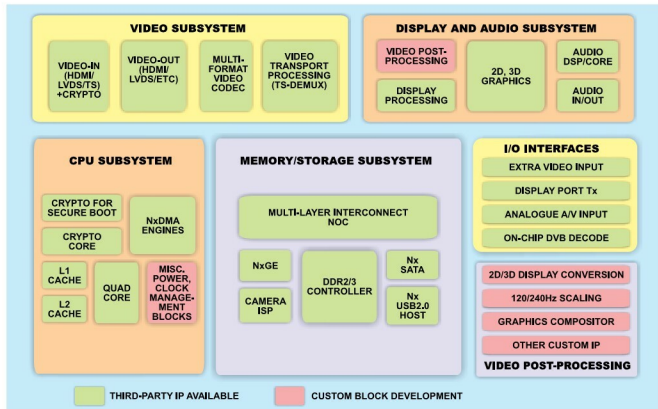
Unique domain-aware modeling-to-validation approach

Unified domain-aware modeling-to-validation approach involves reducing the overall SoC verification and validation cycle time by focusing on reuse and domain-based frameworks during phases of system modeling, verification, rapid prototyping, emulation and silicon validation.

A DTV SoC typically contains

significant processing, hardware-software co-design and dedicated hardware interface blocks. A standard verification environment will not help validate many end-use cases, resulting in a need to move into emulation, SoC architecture modeling and top-level register-transfer level (RTL), and conducting independent verification, validation and emulation of the SoC. However, there are commonalities at each stage.

To arrive at a time- and cost-efficient product development model, one needs to unify these commonalities and build reusable domain-based frameworks. For example, take the scenario of an emulation box where high-level languages cannot be used. Creating a test bench for any stage—verification, validation or prototyping—without the knowledge of the emulation environment requirements would result in rework at the emulation stage. Hence, just like the way OEMs use DTV IP frameworks to arrive at an optimal design, reusable and unified verification approach is



Block diagram of next-generation media SoC